

Melih Ozkurt 76000

Fatih Emir Caylak 87070

Ahmet Bugra Erturk 86877

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Evaluation of Deterministic and Probabilistic Sources for Single Photon Generation

I. INTRODUCTION

Single photon sources are fundamental to the development of quantum information processing as photons can be used to encode, transmit and process information. Since photons propagate at the speed of light and weakly interact with the environment, they are efficient at carrying information, especially over long distances. Therefore, it is essential to develop high quality single photon sources to exploit natural advantages of photons as information carriers. There are two approaches to generate single photons: deterministic sources [1] and probabilistic sources [2]. This distinction refers to internal physical processes rather than overall behaviour, and the boundary between them becomes less clear in practice. Deterministic sources are designed to produce one photon per excitation cycle [1]. The control over timing and the on-demand generation of photons is beneficial for scalable quantum computation and integrated photonic circuits. In contrast, probabilistic sources rely on spontaneous processes, which makes only certain trials usable for applications [2]. Although these sources are simpler to build and are already established in the field of atomic molecular optics, they lack precise photon generation [1, 2].

There are several metrics to evaluate the performance of single photon sources, including but not limited to purity [3], indistinguishability [4] and efficiency [5]. Purity measures the

accuracy of a single photon source to produce exactly one photon without producing multiple photons instead [3]. High purity becomes crucial in some applications of quantum information processing where the algorithm relies on the assumption that the input consists of either zero or one photon. Indistinguishability measures the similarity between produced photons in terms of their properties such as wavelength, polarization and timing [4]. Indistinguishability is necessary for performing effective quantum interference, which is a fundamental process frequently utilized in several fields of quantum information processing. Efficiency quantifies the fraction of generated photons successfully prepared for processing, determining the practical aspect of the source [5]. Deterministic sources typically aim to optimize these metrics simultaneously. However, achieving the best performance concurrently for each of these metrics is challenging due to inherent tradeoffs. In contrast, probabilistic sources generally encounter limitations in efficiency due to their natural dependence on the probability of the photon generation [2]. However, they can relatively achieve high indistinguishability and purity. The viability of the two approaches on single photon generation, deterministic and probabilistic sources, are dependent on how well the source can balance the tradeoff between metrics such as purity, indistinguishability and efficiency, and factors such as scalability and integrability to existing technology.

II. DETERMINISTIC SOURCES

Deterministic photon sources operate by generating a single photon per excitation cycle through controlled emission mechanisms, excluding probabilistic nature of spontaneous emission processes. Among deterministic sources, quantum dot (QD) based sources are the most mature framework for generating high quality single photons [1]. QD based sources perform this by confining information carriers in energy levels, allowing photon emissions when the system is electrically or optically excited. QD setups often utilize optical microcavities since embedding

QDs in those systems enhances emission quality through the Purcell effect [1,4,5,6]. However, this technique is not adequate to build high quality single photon sources since generated indistinguishable photons are dependent on polarization filtering techniques to be collected, which reduces source's efficiency by 50% [5]. Wang et al. addressed this challenge by proposing a method that excites QDs that are embedded in polarization selective Purcell-enhanced microcavities [5]. They illustrated their new method in two different cavity geometry: elliptical micropillars for narrowband excitation and Bragg gratings for broadband excitation. They utilize the "polarization-orthogonal excitation-collection" scheme which is particularly designed to reduce loss caused by polarization filtering. In this scheme, QDs are coupled to one of the cavity modes and collection is performed along the axis orthogonal to previously coupled mode. Ultimately, they achieve high photon indistinguishability (0.975), purity (0.991) and efficiency (0.600) [5].

You and He theoretically improved the polarization engineering approach [4] initially developed by Wang et al. by proposing an asymmetric microcavity design with two orthogonal polarization modes, one broadband for excitation and one narrowband for emission, both resonant with a single QD [5]. They focused on the tradeoff between single photon purity and indistinguishability induced by additional filtering. Similar to Wang et al. 's design [5], You and He's asymmetric design is also capable of preventing %50 efficiency loss caused by polarization filtering [4]. Their simulations show that the scheme improves photon extraction efficiency and coherence without needing external polarization filters. This design also exhibits near-unity purity and indistinguishability, which indicates the method's success in handling the tradeoff between them.

In contrast to the works of Wang et al. and You and He, which focus on improving source efficiency, Tomm et al. emphasize dynamic tunability and integrability of the system by employing gated QDs within open microcavities [4,5,6]. Their gated design stabilizes charge fluctuations and uses electrical tuning to preserve photon coherence at 1 GHz repetition rate with 0.975 indistinguishability [6]. This coherence over long photon sequences is crucial for quantum communication applications, such as entanglement swapping, since it keeps photons identical over long distances allowing interference between photons after a successful transmission, increasing scalability. Furthermore, tunable microcavities can compensate for variation in QD position and emission spectrum, allowing greater control over the system. However, this benefit introduces a demand for spectral alignment, which refers to keeping QD emission resonant with the cavity mode despite the fluctuations. Tomm et al. ensured spectral alignment by adjusting the mirrors using nano-positioners [6]. However, this approach makes the system vulnerable to mechanical vibrations.

Yang et al. solved this challenge by using a piezoelectric actuator to tune the emission wavelength of the QD [1]. They integrated a QD in monolithic Fabry-Perot microcavity with a piezoelectric actuator. This structure can compensate for the mismatch between QD emission and the cavity mode without requiring unstable elements such as nano-positioners from the previous work of Tomm et al. [6]. Therefore, Yang et al.'s compact and stable design reduces the vulnerability to external noise, representing mechanically robust architecture [1]. As a result, their design achieves high photon efficiency (0.949), purity (0.956) and indistinguishability (0.922) [1]. This level of performance in these metrics is essential for scalable quantum information processing as higher accuracy results in less resource overhead.

III. PROBABILISTIC SOURCES

Probabilistic photon sources rely on stochastic (partially unpredictable) processes which produce photons with a probability, where the generation of a single photon is not guaranteed. Probabilistic sources most primarily take advantage of nonlinear optical processes such as spontaneous parametric down-conversion (SPDC) [2]. In SPDC, a strong pump laser is flashed through a nonlinear crystal – commonly periodically poled potassium titanyl phosphate (PPKTP) or lithium niobate (PPLN) – and occasionally splits into a correlated photon pair consisting of “signal” and “idler” photons [2,7]. Detecting the idler photon “heralds” the presence of the other photon, yielding a usable single photon only when the herald fires. The simplicity of the setup and the broad spectrum of wavelengths the process can produce has rendered SPDC a favorable approach to generate photons despite the interactions between the paired photons lowering purity and indistinguishability [7]. High purity is critical to minimize mixedness in the heralded state, ensuring high-fidelity gate operations, while indistinguishability allows higher interference for scalable entanglement schemes. Maximizing these two values remains as the main goal for SPDC sources as with other photon generation schemes.

To improve the purity of heralded single-photons generated through SPDC, Wang et al. present four crystal-engineering techniques: creating a factorable joint spectral amplitude (JSA) without lossy filtering through Gaussian approximation of the phase-matching amplitude (PMA), group velocity matching (GVM), tuning the pump-to-phase-matching bandwidth ratio and joint optimization of pump bandwidth (σ) with crystal length (L) through a local-search algorithm [3]. Their collinear, single-mode simulations in PPKTP (with wavelengths pump = 552 nm, signal = 857.3 nm, idler = 1550 nm) show that an optimization with $\sigma = 1.59$ nm and $L = 11.08$ nm can result in a heralded purity of $p = 0.99967$ under a Gaussian PMA assumption, compared to a purity of 0.969 with a sinc PMA [3]. They derive that this setup reduces the interactions between the

photons, maximizing purity. This breakthrough provides a foundation for next-generation SPDC sources designed for integrated photonic chips, where elimination of bulk filters simplifies the chip's architecture. Utilizing the filterless JSA setup of Wang et al., Lafler & Lanning integrated a segmented SPDC crystal with a sum-frequency-conversion (SFC) stage [3,7]. Their two region architecture combines the PMA of the first SPDC with that of the SFC stage, where the resulting JSA preverses entanglement necessary for polarization. They report a purity of $p > 0.99$ and indistinguishability > 0.98 without external filters, overcoming the purity limitations in bulk PPKTP ($p = 0.80$) [7]. The JSA-shaping techniques in bulk nonlinear crystals Wang et al. propose and Lafler & Lanning implement mirror the role of nanocavities in QD sources with a higher value of purity and indistinguishability compared to Wang et al., resembling the cavity mode selection which narrows the emission spectra, while demonstrating that probabilistic sources that implement internal filtering with no external filtering can outperform deterministic sources [3,5,7].

Although high purity and indistinguishability can be achieved through methods proposed by Wang et al. [3], the conditions where the photons are generated need to be tuned precisely to achieve such high results. Practical integrability of single-photons with pre-existing technology requires compromise of the high values achieved experimentally. Lange et al. embed a PPLN crystal in a cryostat (temperature < 10 K) to shift the wavelength of photon production into the fiber-telecom C-band (1530-1565 nm), allowing telecom integration [8]. They achieve a high heralded purity of $p = 0.983$, with an indistinguishability of 0.663 (comparatively lower than the SPDC source of Lafler & Lanning [7]), limited by imperfect filtering and alignment [8]. Since these limitations are due to the experimental setup and not physical restrictions, Lange et al. theorize the potential for higher indistinguishability, while their photons generated at telecom wavelengths can be integrated directly into existing fiber-optic infrastructures without external

wavelength conversions and the cryogenic conditions allow the implementation of superconductor nanowire technologies (which only operate at extremely low temperatures) used in advanced QD platforms (such as monolithic Fabry-Perot cavities that Yang et al. implement [1]), paving the way for potential hybrid systems of probabilistic and deterministic sources [8].

The superconducting nanowires are also implemented into the setup of Davis et al., who combine a bulk SPDC source with a photon-number-resolving superconducting nanowire (PNR-SNSPD), which separates the instances of single-pair photons and multi-pair photons [2]. By actively suppressing multiphoton heralds, they boost heralding efficiency by 25% without sacrificing purity, achieving a purity of $p = 0.993$ [2]. The PNR-SNSPD method which actively filters photon generation is an alternative to the setup of Lafler & Lanning's JSA engineering, which suppresses side-lobes (unwanted correlations between signal and idler photons) passively [7]. Davis et al. demonstrate that real-time detection can eliminate residual errors (multi-pair photons), a paradigm that is also explored in QD sources through resonant excitation gating and fast optical switches [2,6]. The studies of Davis et al. and Lafler & Lanning together illustrate a potential for a hybrid approach combining passive JSA shaping and active multiphoton suppression aiming for high performance on demand [2,7].

IV. CONCLUSION

The development of single photon sources present several challenges, which include achieving high purity, indistinguishability, and integrability. Both the deterministic approach and the probabilistic approach have an excess of strengths and weaknesses when measured against the listed metrics. Deterministic sources such as quantum dots need to be excited and the light interaction needs to be controlled along with electrical cavity stimulation to allow for an acceptably pure signal. However, the fabrication and maintenance of the parts required are a significant issue

that limits their practical application. The other approach, which is probabilistic sources such as spontaneous parametric down-conversion, offer simple implementations and compatibility with existing fiber infrastructures through techniques such as factorable joint spectral amplitudes and photon-number-resolving detection and can achieve comparable or even superior purity and indistinguishability to deterministic systems. However, their stochastic nature limits their efficiency. With these advantages and drawbacks, the development of both approaches in single photon generation contribute to the field of quantum technology by providing a flexible framework for specific implementations. Future research, with potentially hybrid models that integrate both approaches, will most likely focus on optimizing the quality of single photons while utilizing deterministic and probabilistic aspects of existing setups. Therefore, it is essential to conduct further research on both deterministic and probabilistic photon sources.

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